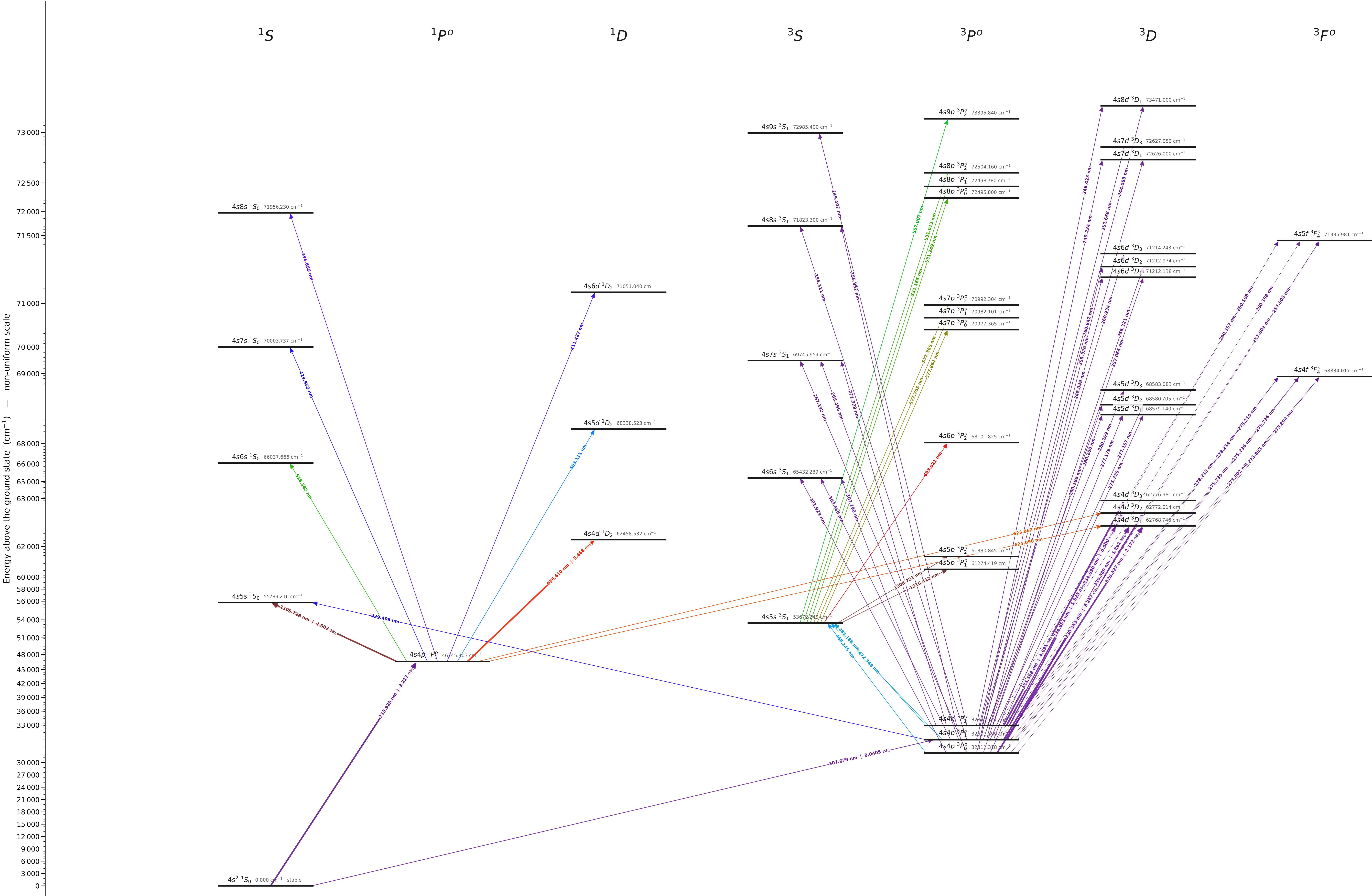
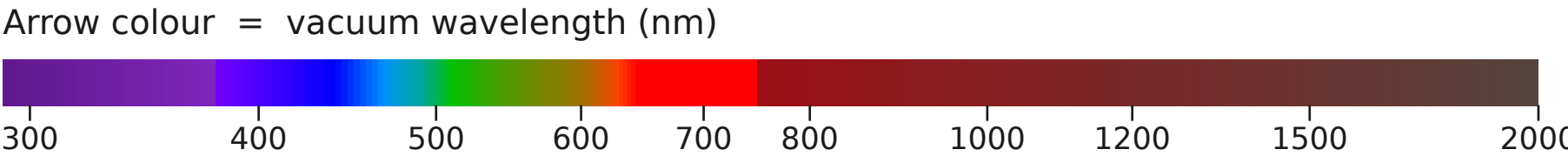


Zinc

Zn (Zn I). Strongest classified transitions below 2 μm (NIST): 45 levels, 70 lines.

Zn⁺ ionisation limit 75769.31 cm⁻¹ = 9.39420 eV (131.980 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||U'|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U er U'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
4s ² 1S ₀ - 4s4p 1P ₁ ^o	213.9248	213.8574	1401.3919	3.217	7.140e+08	-	-	
4s ² 1S ₀ - 4s4p 3P ₁ ^o	307.6791	307.5897	974.3674	0.040	3.800e+04	-	-	
4s4p 3P ₀ ^o - 4s4d 3D ₁	328.3271	328.2326	913.0907	2.172	9.000e+07	-	-	
4s4p 3P ₂ ^o - 4s4d 3D ₂	330.3534	330.2583	907.4902	3.267	1.200e+08	-	-	
4s4p 3P ₂ ^o - 4s4d 3D ₁	330.3890	330.2939	907.3922	1.891	6.700e+07	-	-	
4s4p 3P ₂ ^o - 4s4d 3D ₃	334.5975	334.5013	895.9793	4.691	1.700e+08	-	-	
4s4p 3P ₂ ^o - 4s4d 3D ₂	334.6531	334.5569	895.8305	1.923	4.000e+07	-	-	
4s4p 3P ₂ ^o - 4s4d 3D ₁	334.6897	334.5935	895.7325	0.500	4.500e+06	-	-	
4s4p 1P ₁ ^o - 4s4d 1D ₂	636.4105	636.2346	471.0678	5.468	4.700e+07	-	-	
4s4p 1P ₁ ^o - 4s5s 1S ₀	1105.7283	1105.4256	271.1267	4.002	2.400e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).